

NICOLA CARBONI

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Academic Appointments

Assistant Professor <i>University of Illinois Urbana-Champaign</i> School of Information Sciences	Mar 2025 –
Lecturer <i>University of Geneva</i> Digital Humanities – Faculty of Arts & Humanities	Aug 2022 – Aug 2024
Postdoctoral Fellow <i>University of Geneva</i> Digital Humanities – Faculty of Arts & Humanities	Feb 2021 – Mar 2024
Research Fellow in Semantic Architecture <i>University of Zurich</i> Swiss Art Research Infrastructure – Department of Art History	July 2018 – Dec 2020
Research Fellow <i>IUAV University of Venice</i> ERC Project – Burckhardt Source, Centro Studi ClassicA	Jan 2019 – Sept 2019
Digital Humanities Fellow <i>Harvard University</i> Harvard Center for Italian Renaissance Studies	Feb 2018 – June 2018
Marie Curie Early-Stage Researcher <i>French National Centre for Scientific Research</i> Models and Simulations for Architecture and Heritage Laboratory	June 2014 – July 2017

Education

PhD, School of Engineering National Technical University of Athens <i>Thesis:</i> Conceptualization and Semantic Description in Digitized Tangible/Intangible Content	Jun 2014 – July 2020
International Master in Digital Library Learning Jointly awarded by the Universities of Oslo, Tallinn, Parma	Sept 2011 – Oct 2013
Bachelor in Cultural Heritage Studies Università di Pisa	Sept 2007 – Sept 2010

Research Visits

3DOM, Fondazione Bruno Kessler (FBK) <i>Host: Fabio Remondino</i>	Feb 2016
KAAC, German Archeological Institute (DAI) <i>Host: Jennifer von Schwerin</i>	Feb 2015
FORTH ICS <i>Host: Martin Doerr</i>	Jan 2015

Funding

TextEnt – TEXTual ENTities in time Swiss National Science Foundation – SPARK Co-PI: Simon Gabay. Amount: 100,000 CHF	2024
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Teaching

University of Illinois Urbana-Champaign IS 477 – Data Management, Curation & Reproducibility	2025 –
University of Geneva Digital Art History	2023–24
Les données des GLAM: de l'analyse à la curation numérique	2023–24
Images Numériques I & II	2022–24
Knowledge Graph et Représentation des Connaissances	2022–23
Biais dans les collections numériques: données et représentation(s)	2021–23
Représentation des connaissances	2021–22

Publications

Journals

- 2025 Bruseker, G., **Carboni, N.**, Fielding, M., Nenova, D., Haensli, T. *The Semantic Reference Data Modelling Method: Creating Understandable, Reusable and Sustainable Semantic Data Models*. *Journal of Open Humanities Data*, 11.
- 2024 Joyeux-Prunel, B., **Carboni, N.**, Barras, M. *Plotting the Geopolitics of 20th-c. Modern and Avant-Garde Illustrated Periodicals*. *Journal of European Periodical Studies*, 9(1).
- 2024 Joyeux-Prunel, B., Dossin, C., **Carboni, N.** *Style Revisited*. *Artl@s Bulletin*, 13(2).
- 2024 Joyeux-Prunel, B., Barras, M., **Carboni, N.** *Greater Aura? The 19th and 20th Century Circulation of Art Reproductions in the Press through a Digital and Historical Perspective*. *Život umjetnosti*, 114(1).
- 2023 **Carboni, N.** *The Mediatization of the Early Automobile: A Visual Analysis of the Illustrated Press in the late 19th and Early 20th century*. *Artl@s Bulletin*, 12(1).
- 2023 Joyeux-Prunel, B., Barras, M., **Carboni, N.** *Une Europe par les arts? Les périodiques illustrés au-delà du Musée imaginaire*. *Artl@s Bulletin*, 12(1).
- 2023 Joyeux-Prunel, B., **Carboni, N.**, Jeanrenaud, A., Viaccoz, C., Belina, C., Gauffroy-Naudin, T., Barras, M. *Un œil mondial? La mondialisation par l'image au prisme du numérique – le cas du projet Visual Contagions*. *Sociétés et représentations*, 55.
- 2021 **Carboni, N.**, Federici, A., Ciriogliaro, G. *Tacita Dean's Buon Fresco: developing a digital Atlas on The life of Saint Francis Fresco cycle in the upper Basilica of Saint Francis of Assisi*. *Linguistica e Letteratura*, XLVI(1/2).
- 2021 Mirabaud, S., Stoleru, A., Bosc-Tiessé, C., **Carboni, N.**, Guéna, F., Messaoudi, T. *Vers la formalisation d'un modèle ontologique pour la description des églises rupestres peintes d'Éthiopie*. *Histoire de l'art*, 87.
- 2019 **Carboni, N.**, De Luca, L. *An Ontological Approach to the Description of Visual and Iconographical Representations*. *Heritage*, 2(2), 1191–1210.
- 2017 Bruseker, G., **Carboni, N.**, Guillem, A. *The Role of Formal Ontology and CIDOC CRM*. *Heritage and Archaeology in the Digital Age*, pp. 93–131. Springer.
- 2017 **Carboni, N.**, De Luca, L. *Towards a Semantic Documentation of Heritage Objects through Visual and Iconographical Representations*. *International Information Library Review*, 49(3).
- 2016 **Carboni, N.**, De Luca, L. *Towards a conceptual foundation for documenting tangible and intangible elements of a cultural object*. *Digital Applications in Archaeology and Cultural Heritage*, 3(4):108–116.
- 2016 **Carboni, N.** et al. *Data Provenance in Photogrammetry Through Documentation Protocols*. *ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences*, III-5:57–64.
- 2015 Bruseker, G., Guillem, A., **Carboni, N.** *Semantically Documenting Virtual Reconstruction: Building a Path to Knowledge Provenance*. *ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences*, II-5/W3:33–40.

Conferences

- 2026 **Carboni, N.** *Mediating Access to Art-Historical Data using LLMs and RAGs*. In *114th College Art Association Conference*, Chicago, United States.
- 2026 **Carboni, N.** *Exploring Cultural Knowledge Graph Data using Large Language Models and Retrieval-Augmented Generation*. In *The Art Museum in the Digital Age 2026*, Vienna, Austria.
- 2026 Romanello, M., Gabay, S., **Carboni, N.** *Predicting the Fictional Time and Space of French Theatre Plays by Using Large Language Models*. In *2025 IEEE International Conference on Cyber Humanities*, Florence, Italy.
- 2026 Gabay, S., Kuenzli, P., Rubino, R., Skura, G., De Biasi, L., Falcone, J.-L., Charpillot, C., Romanello, M., Clérice, T., Janès, J., Ing, L., **Carboni, N.** *Computational challenges for the massive production of enriched pre-editions*. In *2025 IEEE International Conference on Cyber Humanities*, Florence, Italy.
- 2025 **Carboni, N.** *Poster – A Knowledge Graph-enhanced RAG framework for Cultural Heritage Data*. In *K-CAP 2025*, Dayton, United States.
- 2025 **Carboni, N.** *Querying Art Historical Knowledge Graphs: a Natural Language Interface for Exploring CIDOC-CRM Data Using LLM*. In *Digital Publishing for Art History*, Rome, Italy.
- 2025 **Carboni, N.** *AI-mediated interfaces for exploring cultural knowledge repositories*. In *Ctrl+S Culture: AI and Heritage in a Digital World*.

- 2025 Barria Bignotti, C. F., **Carboni, N.**, Barras, M. *Geneva Cultural Heritage Research Platform*. In *The Art Museum in the Digital Age 2025*, Vienna, Austria.
- 2024 **Carboni, N.** *Ontological patterns for modeling the validity of spatiotemporal statements*. In SWODCH'24, Tours, France.
- 2024 **Carboni, N.** *Integration and Analysis of Exhibition Information*. In CIDOC 2024, Amsterdam, The Netherlands.
- 2024 **Carboni, N.** *Ontological patterns for modeling art exhibitions*. In 8th International Conference on Metadata and Semantics Research, Athens, Greece.
- 2024 **Carboni, N.** *Visual Exchanges as a Network: The Case of Avant-Garde Periodicals*. In *The Historical Network Research Conference 2024*, Lausanne, Switzerland.
- 2024 **Carboni, N.** *The structures of visual exchanges*. In *Reimagining Annotation for Multimodal Cultural Heritage*, Rennes, France.
- 2023 **Carboni, N.**, Usel, T., Joyeux-Prunel, B. *Linked Open Data pour l'intégration d'informations spatio-temporelles sur les expositions*. In *Humanistica 2023*, Geneva, Switzerland.
- 2023 **Carboni, N.**, Barras, M., Joyeux-Prunel, B. *Pister des circulations visuelles à l'échelle mondiale*. In *Humanistica 2023*, Geneva, Switzerland.
- 2023 **Carboni, N.** *A computational analysis of image globalization: the impact of images in the periodical press*. In *Seventh European Congress on World and Global History 2023*, Den Haag, Netherlands.
- 2023 **Carboni, N.** *Analysis of the Material Limitations of Data Infrastructures: Overcoming the Database*. In *From Hype to Reality: AI in the Study of Art and Culture*, Zurich, Switzerland.
- 2023 **Carboni, N.**, Joyeux-Prunel, B., Barras, M. *A Computational Analysis of the Transmission and Impact of Images in the Periodical Press*. In *College Art Association Conference*, New York, United States.
- 2023 Usel, T., **Carboni, N.**, Joyeux-Prunel, B. *Linked Open Data for Horizontal Integration of Exhibition Information*. In *The Art Museum in the Digital Age*, Vienna, Austria.
- 2022 **Carboni, N.**, Joyeux-Prunel, B. *Images in Global Circulation: A Multi-Scalar Approach*. In *Digital Art History IV*, Zagreb, Croatia.
- 2022 **Carboni, N.**, Federici, A., Cirnigliaro, G. *Tacita Dean's Buon Fresco*. In *Renaissance Society of America*, Dublin, Ireland.
- 2022 **Carboni, N.** *The Rise of the Machine*. In *Thinking Europe Visually*, École normale Supérieure de Paris, France.
- 2022 **Carboni, N.** *Exploring the Visual with Formal Ontologies*. In *5th Swiss Congress for Art History*, University of Zurich, Switzerland.
- 2022 **Carboni, N.**, Ghelardi, M. *Textual datification: the case of BurckhardtSource*. In *36th Congress of German Art Historians 2022*, University of Stuttgart, Germany.
- 2020 **Carboni, N.**, Bruseker, G. *Representation for reference entity-types: introducing the Semantic Reference Data Models*. In *CIDOC Conference*, Geneva, Switzerland.
- 2019 **Carboni, N.** *Using Semantic Web Technologies for the Classification of Tangible and Intangible Elements of an Iconographical Representation*. In *4th Swiss Congress for Art History*, Mendrisio, Switzerland.
- 2016 Ramos, M., Bellido, D., **Carboni, N.**, Domajnko, M., Stathopoulou, E., Stavropoulou, G., Morabito, D., Remondino, F. *Complex 3D heritage architectures accessible on the web*. In *ARQUEOLÓGICA 2.0*, Valencia, Spain.

Workshops & Seminars

- 2025 **Carboni, N.** *AI and Knowledge Graphs: From Datafication to Accessibility*. In *Cultural Heritage and AI*, Leiden, Netherlands.
- 2025 **Carboni, N.** *Integrating Computational and Historical Methods for investigating Image Circulations*. In *Encoding the Image*, Toronto, Canada.
- 2024 **Carboni, N.** *Ontological Patterns for Integrating Art Exhibitions Datasets*. In *Documenting Exhibitions with the Semantic Web*, Montreal, Canada.
- 2024 **Carboni, N. Keynote:** *Contextualize Heritage 3D Objects using Semantic Annotations: from Provenance to Interpretation(s)*. In *Advanced 3D Archaeological Documentation and Linked Open Data*, Lund, Sweden.
- 2023 **Carboni, N.** *A data driven approach to the study of car imagery*. In *Representing Automobility in Literature and Film*, Frankfurt, Germany.

- 2023 **Carboni, N.**, Joyeux-Prunel, B. *Integration and analysis of exhibition information using Linked Open Data*. In *Semantic Web, Cultural Heritage, and Art Historical Knowledge*, Malaga, Spain.
- 2023 **Carboni, N.** *The Mechanics of Image Study: Analysing Image Circulation across Large Corpora of Visual Data*. In *Woodcuts. Studying the Images of Popular Prints*, Valencia, Spain.
- 2021 **Carboni, N.** *An Ontology for the Description of Visual Heritage*. In *TORCH, The Oxford Research Centre in the Humanities*, Oxford, United Kingdom.
- 2019 **Carboni, N.**, Muller, S., Ghelardi, M. *Towards a Foundation for Documenting Digital Textual Resources with CIDOC-CRM*. In *Scholarly Digital Editions, Graph Data-Models and Semantic Web Technologies*, Lausanne, Switzerland.
- 2017 Alliez, P., Bergerot, L., Bernard, J.-F., Boust, C., Bruseker, G., **Carboni, N.** et al. *Digital 3D objects in art and humanities: Challenges of creation, interoperability and preservation*. In *PARTHENOS*.
- 2017 **Carboni, N.** *Description of linked data and "historical metadata" in 3D models*. In *PARTHENOS 2017*, Bordeaux, France.
- 2015 **Carboni, N.**, De Luca, L. *Interlinking physical and conceptual dimensions of heritage artefacts into semantically-enriched representations*. In *Museums as Cultural, Urban, Creative and Learning Interface*, Marseille, France.

Round Tables

- 2023 **Carboni, N.** *Analyse Des Données Et Musées*. In *Ouvrir Les Musées Au(x) Numérique(s)*, Humanistica 2023, Geneva, Switzerland.
- 2022 **Carboni, N.** *SPARQL for Visual Sources*. In *Beyond Linked Data: Data Querying Services through API and SPARQL*, Renaissance Society of America, Dublin, Ireland.
- 2021 **Carboni, N.** *Semantic Description of 3D Sources*. In *Early Modern Material Culture: New Approaches in Digital Humanities II*, Renaissance Society of America, Virtual.

Invited Lectures

- 2025 *Discovering connections: documentation of visual heritage using graph technologies*. Digital Heritage Association (DHA), Indiana University, Bloomington.
- 2023 *Temporal Uncertainties*. RDF & Linked Data Seminar, University of Geneva.
- 2022 *Images in Global Circulation: A Multi-Scalar Approach*. Data Science Day, University of Geneva.
- 2022 *A data-driven approach to the study of images circulation*. Digital History – Data for History Lectures series, Humboldt University of Berlin.
- 2022 *Historical semantic data: challenge and possibilities*. Digital Humanities PhD Seminar, University of Geneva.
- 2021 *Using formal ontologies to explore visual culture(s)*. Digital Humanities PhD Seminar, EPFL, Lausanne.
- 2021 *Documenting heritage*. Research Seminar Art in the making, Institut national d'histoire de l'art, Paris.
- 2019 *Digital Art History*. Summer school Stucchi E Stuccatori Ticinesi a Roma, Rome.
- 2019 *Linked data for Digital Humanities*. Master Comunicazione Storica, Rome.

Organization

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| Roundtable | Ouvrir Les Musées Au(x) Numérique(s) – Humanistica 2023, Geneva |
| Conference | Humanistica 2023, Geneva |
| Seminar | Conversation Series in Digital Art History. Artl@s/Visual Contagions Research Seminar |
| Roundtable | Data Sustainability II – Renaissance Society of America Virtual 2022 |
| Seminar | Styles Revisited: From Iconology to Digital Image Studies (2021–22, 2022–23) |
| Conference | ITN-DCH – Digital Heritage 2017 Final Conference, Olimje, Slovenia |
| Workshop | Workflow for the Integration of Heritage Digital Resources, Olimje, Slovenia |

Collaborations

- 2023 - 2025 Development of Semantic Data Catalogues for the MAH Collection, Musée d'Art et d'Histoire, Geneva

Exhibitions

Digital Curation and Coordination – *Visual Contagions. Images in Globalization*. Jeu de Paume Museum, Paris

Digital Curation and Coordination – *The images that shaped Europe*. Europeana Foundation, Den Haag

Direction

Co-director Artl@s project

Co-director TextEnt

Service

Member Editorial Advisory Board – Digital Library Perspectives

Reviewer Artl@s Bulletin, Digital Library Perspectives, ACM Journal on Computing and Cultural Heritage, Journal of Cultural Heritage, Heritage, Studies in Digital Heritage, Journal of Open Humanities Data, Semantic Web journal, Routledge's Museum and Heritage Studies

Memberships

International Council of Museums (ICOM) • CIDOC-CRM Special Interest Group • Digital Art History Society • College Art Association (CAA) • Association for Computers and the Humanities (ACH)

Languages

Italian Native

English Fluent

French Advanced